

Project Diary Seenu

Weeks 1 to 10

Field	Details
Student	Seenu (A2345923074)
Role	Research and evaluation
Project	Project MAESTRO - Design and Evaluation of a Safe Multi-Agent System for Natural Language Driven Desktop Task Automation
Guide	Dr. Rajni Sehgal Kaushik
Group	Group 298

Diary Coverage

This diary records my contribution to the Minor Project across Weeks 1 to 10. The first part covers the original Week 1 to Week 5 work. The second part is separated as the continuation work for Weeks 6 to 10.

Part One Weeks 1 to 5

Week 1

The first week was mainly about finding out what field we were standing in. Before any paper could be read sensibly we had to know which areas the project touches, and I worked through that until we had five: artificial intelligence, natural language processing, multi-agent systems, desktop automation and AI safety. Naming them properly mattered, because each one has its own vocabulary and its own set of venues, and searching without knowing that wastes a lot of time. I ran the first searches on Google Scholar and arXiv to see how much work exists in each area and how recent it is, and it was clear immediately that the desktop agent work is very new and moving quickly. I also did the economic side of the feasibility study, listing every tool we intend to use along with its licence and its free limits, which is how we can honestly write zero against the bill of material.

Week 2

This was the reading week. I collected around thirty papers and, rather than reading them in the order I found them, I set up a matrix first with columns for the year, the approach to planning, whether the system executes on a real machine, what benchmark it uses, and what it reports. Filling a table forces you to notice what a paper leaves out, which turned out to be the most useful decision I made this month. On the agent side I read the work on reasoning combined with acting, on task planning with tools, on open-ended agents that learn skills for themselves, and on frameworks where several specialised agents cooperate instead of one large prompt doing everything. From all of that I wrote the first part of the Literature Review chapter, organised by theme rather than one paragraph per paper, because a list of summaries is not a review.

Week 3

This week the study moved to the systems people actually use, and I ran the comparative study while Jairaj tested the same three systems on the machine. I compared them property by property rather than feature by feature, and the properties I chose are the ones that matter for our argument: how an action is represented before it runs, whether there is any classification of danger, whether the user gets a preview, whether anything can be undone, whether there is a real audit trail, and whether the system defends itself against instructions hidden inside a file or a web page. The pattern was the same in every column. Capability is strong and improving. Everything to do with

safety is either missing or is a dialog box. The finding I keep coming back to is the last row of my table, which is that none of these systems report a safety measurement at all, so there is no way to say that one is safer than another.

Week 4

I wrote the research gap chapter this week, and I was careful to make every claim in it something a reader can check against my own matrix rather than something they have to accept on trust. The gap came out in four parts. Existing work measures how capable an agent is and not how safe it is. A confirmation popup is an interface feature with no testable properties, whereas a typed plan with fixed scoring rules can actually be examined. Resistance to instructions hidden in documents is never reported as a headline number for desktop systems. And there is no public dataset of instructions paired with plans for this task, least of all one carrying safety labels. Writing the gap also pushed me into thinking about measurement early, because if we are going to claim our system is safer then the word has to mean a number, and I began sketching what those numbers should be.

Week 5

The requirement work this week needed real situations rather than a wish list, so I wrote the user scenario analysis. I built three personas, the student who is not a programmer and whose files are scattered everywhere, the developer who wants speed but also wants a log he can check afterwards, and the evaluator who is looking for method and honest numbers rather than a polished demo. Then I wrote out the situations each of them would actually put to the system, which is what produced the six categories of task the requirements now cover. In parallel I started defining the measurements properly, and the important one is the unsafe execution rate, which counts how often a consequential action ran without the confirmation it should have had. Our target for it is zero. I also decided it must always be reported together with the rate at which the system asks for confirmation unnecessarily, because a system that asks about everything would score zero on the first number while being useless.

Where I stood after Week 5

The literature review, the comparative study and the research gap are done and written, and five weekly reports are filed. What comes next is mine as well: the evaluation methodology chapter, where the hundred benchmark tasks are designed and each measurement is defined formally, followed by the seed dataset with its annotation guidelines and an agreement score calculated on a shared portion so that the labelling can be shown to be consistent.

Part Two Weeks 6 to 10 Continuation Work

Week 6

This week I worked on turning the research gap into language that can be measured. While the Action IR and risk model were being frozen, I checked that the definitions would support evaluation later. I focused on the difference between an action that is only inconvenient and an action that is unsafe, because the project cannot defend a safety claim unless the boundary is written clearly. I helped review the R0 to R3 tiers, the hard block list and the trust model, and I made notes on how each rule could later become a test or a metric.

Week 7

My work in Week 7 was tied to the architecture chapter. I reviewed the component responsibilities from the point of view of evaluation: what the NLP layer promises, what the planner is allowed to do, where the safety layer begins, and which part of the system is responsible for proof that an action succeeded. I also helped write the data model and sequence descriptions so that denial, failure, rollback and injection blocking are visible as normal flows rather than exceptions hidden in the code.

Week 8

During the diagram and design consolidation week I checked the diagrams for consistency with the safety argument. I paid special attention to the trust boundary diagrams because they explain why file and web content are treated as untrusted data. I also helped prepare the algorithm descriptions for risk scoring, dry run preview, DAG scheduling and injection filtering. This made the design easier to defend because the algorithms are described step by step rather than only as component names.

Week 9

Week 9 was mainly my evaluation methodology week. I worked on the 100 task benchmark structure, the grouping of tasks by category and difficulty, and the definitions of the main metrics. I wrote the logic for reading unsafe execution rate together with false confirmation rate, because a system that asks confirmation for everything is not actually usable. I also helped fix the baselines and ablations so that the final report can explain what changes when the safety layer, dry run, postconditions, critic or trained intent model are removed.

Week 10

This week I worked on the dataset specification and quality checks. The dataset needed realistic instructions, not just clean textbook commands, so I helped review ambiguous and adversarial examples as well as ordinary file, browser, system and drafting tasks. I also checked that unsafe and refusal examples are represented, because a model that never sees refusal examples will learn compliance by omission. By the end of the week the dataset, benchmark and evaluation plan were connected to the research questions.

Present Position After Week 10

By the end of Week 10 my main contribution is the evaluation side of the project. I helped convert the research gap into metrics, benchmark categories, adversarial cases and dataset quality rules. My immediate remaining work is to support the final report assembly, make sure the evaluation claims are quoted with the correct caveats, and prepare the viva answers around why MAESTRO measures safety rather than simply claiming it.